ZenoEnergy: Verifier-Preserving Learned Candidate Ordering for UPBA v2 Settlement Search

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Abstract

ZenoEnergy studies whether learned energy scorers can reduce candidate search cost for UPBA v2 partial-fill exact-in settlement search while preserving deterministic verification. The system follows the rule: model proposes, verifier decides. The first stable checkpoint was a 97-parameter CPU-only linear ranker trained on synthetic, verifier-labeled bounded candidate sets. The current preferred research checkpoint is `gemini_mlp_v6_seed20260519`, a 6,273-parameter pure-Python MLP. On a held-out synthetic corpus with 1,983 winner-bearing batches and approximately 20 candidates per batch, v6 places the exact verifier winner first in 99.75 percent of batches and in the top 5 and top 10 in 100.0 percent of batches. Mean winner position falls from 19.995 under exhaustive ordering to 1.0025 under v6 learned ordering, while invalid accepts remain zero. The paper records the empirical result, the formal safety boundary, the blockers to production readiness, and a transfer path for applying the same advisory-energy pattern to AutoTrader candidate planning. The current production-adjacent path is an evidence bundle rather than a release claim: source-manifested and coverage-profiled real replay reports are assembled and passed through a fail-closed advisory ranking promotion gate. The latest leaderboard compares seven UPBA v2 advisory rankers and promotes v6 as the current research checkpoint. The AutoTraderEnergy hard synthetic cross-seed receipt records learned mean guard calls of 1.010 versus 4.312 for hand energy, with zero invalid accepts; real shadow replay remains the next AutoTrader milestone. The checked AutoTrader refiner records a 12.00 mean synthetic objective gain across 160 contexts with zero policy-invalid selections; the preconditioned refiner replay improves this to 13.01. The JEPA/ZenoLogic boundary receipt records future-tension scoring and the hazard that energy negation can invert hard safety barriers. The source-level AutoTrader JEPA UX receipt now records future-risk and UX evidence: later-policy-failure AUC 0.8144, slippage, budget, and drawdown stress correlations of 0.6133, 0.5592, and 0.5556, zero invalid accepts over 96 generated contexts, and explicit fields showing that neither the model nor the UX card authorizes execution.

Keywords: energy-based ranking, deterministic verification, DeFi settlement, candidate search, bounded optimality, UPBA, AutoTrader, advisory machine learning.

1 Introduction

ZenoDEX settlement search is a finite candidate-selection problem once a pool, intent set, price grid, and fill-vector family have been fixed. The verifier can check a submitted certificate deterministically, yet exhaustive candidate ordering may waste verifier calls when the winning candidate appears late in the candidate list. ZenoEnergy v0 asks a narrow empirical question:

> Can a small learned energy scorer reduce UPBA v2 candidate-search cost while leaving deterministic certificate verification unchanged?

The answer on the current bounded synthetic benchmark is affirmative. The learned score is useful because it moves high-quality candidates earlier in the verification schedule. It does not establish settlement validity, candidate set completeness, bounded-grid optimality, or permission to stop early without a deterministic suffix certificate.

2 Problem Statement

For a settlement context x, let $C(x)$ be a finite candidate set, let $V_x(c)$ be the deterministic verifier result for candidate c, and let $J_x(c)$ be the deterministic settlement objective ordered lexicographically by executed volume and surplus. The exhaustive verifier winner is

$$c^\star(x) := \arg\max\{J_x(c) \mid c \in C(x),\ V_x(c) = \text{accept}\}.$$

ZenoEnergy learns a scalar energy

$$E_\theta(x, c) \to \mathbb{R}.$$

Lower energy means earlier verification priority. The optimization target is the expected position of $c^\star(x)$ after sorting by energy:

$$\min_\theta\ \mathbb{E}_x[\text{position}_{\text{sort}(E_\theta)}(c^\star(x))].$$

The settlement rule is unchanged:

$$\text{accepted}(c) := V_x(c).$$

3 Safety Boundary

The implementation keeps the energy module outside the consensus-critical trusted computing base. The dependency direction is one-way:

$$\texttt{src.energy} \to \texttt{src.core.uniform\_batch\_clearing}.$$

Verifier modules do not import `src.energy`. Energy code may import verifier APIs for offline labeling, benchmark construction, and tests.

The core safety property is:

$$\text{LowEnergy}_\theta(x, c) \wedge V_x(c) = \text{reject} \Rightarrow \text{SettlementRejected}.$$

Thus an attractive model score can change order only. It cannot authorize a settlement, mutate ledger state, enter a state root, modify replay, or replace certificate verification.

4 Relation to Energy-Based Reasoning

ZenoEnergy follows the structured-output energy-based learning lineage developed by LeCun and collaborators. In that formulation, an EBM assigns a scalar energy to configurations of observed and predicted variables; inference searches for low-energy configurations, and training shapes the energy surface so desired configurations have lower energy than undesired configurations [2]. LeCun and Huang also frame discriminative EBM training around loss functions that can compare candidate configurations without requiring the full probabilistic normalization cost [1].

Song and Kingma describe classical energy-based models as non-normalized models whose probability density or mass is specified up to an unknown normalizing constant, which makes likelihood training difficult and motivates alternatives such as score matching and noise-contrastive estimation [4]. ZenoEnergy adopts the discriminative, structured-output part of this lineage. It uses an energy as a ranking function:

$$E_\theta(\text{context}, \text{candidate}) \to \text{scalar}.$$

This is closer to the reasoning-energy framing in which an energy score is used to evaluate partial states, constraints, objectives, and progress during planning [5]. It is also compatible with LeCun's broader view of EBMs as scalar compatibility functions for predictive world models and planning under uncertainty [3]. In ZenoEnergy, the partial state is a candidate settlement, the constraints are verifier obligations, and the objective is early discovery of the deterministic winner.

5 Model and Training

The retained linear baseline is a ranker with 96 feature weights and one bias:

$$E_\theta(x, c) = w^\top \phi(x, c) + b.$$

It has 97 parameters and runs without PyTorch. The promoted v6 research checkpoint uses a pure-Python MLP with shape $96 \to 64 \to 1$ and 6,273 parameters. The MLP remains an advisory ranker. Its output cannot authorize settlement, mutate state, enter a state root, or replace deterministic verification.

Training uses pairwise hinge ranking over candidates from the same batch:

$$L = \max(0,\ m + E_\theta(\text{good}) - E_\theta(\text{bad})).$$

The preferred linear checkpoint uses weighted pair updates:

$$\begin{aligned}\text{pairWeight} = \mathbf{1}_{\text{winnerPair}}\lambda_w &+ \lambda_v \cdot \text{normalizedVolumeGap} \\ &+ \lambda_s \cdot \text{normalizedSurplusGapWhenVolumeTies},\end{aligned}$$

clipped to a fixed maximum. This places more training pressure on the remaining hard cases: valid candidates ranked ahead of slightly better valid winners. The corrected Gemini evaluation path applies each model's declared feature surface before scoring; streaming cross-seed and hard-case evidence now uses that same adapter.

6 Synthetic Candidate Generation

The synthetic generator creates single-pool CPMM exact-in UPBA v2 batches. It samples reserves, fees, users, balances, exact-in intents, candidate price ratios, and deterministic fill vectors. Candidate families include valid candidates, suboptimal valid candidates, balance failures, limit-price violations, negative-reserve cases, invariant failures, noncanonical fill vectors, all-zero candidates, random noisy candidates, attractive output mismatches, unreduced price ratios, and schema/policy mismatches.

Every candidate is labeled by deterministic verifier execution. Each row stores the 96-dimensional feature vector, candidate hash, generator provenance, verifier validity, objective volume, objective surplus, hand energy, target energy, and winner label.

7 Formal Contracts

The Lean development records three layers of safety.

Exact generation. Generated data supports a mathematical optimality claim only when the candidate corpus is sound and complete for the bounded family:

$$\text{GeneratedCorpusExact}(G, F) := \text{CompleteAuditSet}(G, F) \wedge \text{SoundAuditSet}(G, F).$$

Advisory reordering. If ranked order is a permutation of an exact candidate set, a deterministic verifier certificate over the reordered list proves the same bounded global weak optimum. The corresponding Lean theorem is recorded in `UniformBatchOptimality.lean` as the advisory-reordering certificate theorem.

Full fallback. Full fallback is exhaustive-equivalent when the checked order is a permutation of the original finite candidate set:

$$\text{FullFallbackEquivalentOrder}(C, O) := O.\text{Perm}(C).$$

The theorem `full_fallback_equivalent_order_preserves_weak_optimality_iff` states:

$$\text{WeaklyOptimalIn}(w, O) \iff \text{WeaklyOptimalIn}(w, C).$$

Early stop. Stopping before full fallback requires more than a low energy score. The Lean definition `CheckedStopCertificate` requires the current winner to be in the checked set and to weakly dominate both checked candidates and the unchecked suffix. The exact-full checked-stop theorem lifts that certificate to global weak optimality when the full candidate set is exact and the checked prefix plus suffix is a permutation of the full list.

8 Experimental Results

The stored training corpus contains 199,860 candidate rows from 10,000 synthetic batches. The holdout corpus contains 39,979 candidate rows from 2,000 synthetic batches, with 1,983 batches containing at least one verifier-valid candidate. Metrics below are computed over those 1,983 winner-bearing batches.

Table 1: Current UPBA v2 advisory model leaderboard.

Model	Holdout	Top-1	Cross	Worst top-1	Hard top-1	Misses
Gemini MLP v6	1.0025	0.9975	1.0036	0.9839	0.9940	9
Gemini highwinner linear	1.0066	0.9934	1.0076	0.9839	0.9919	12
Gap-weighted linear	1.0166	0.9834	1.0175	0.9677	0.9854	65
Gemini linear v5	1.0217	0.9788	1.0202	0.9637	0.9839	24

The v6 checkpoint is the current preferred UPBA v2 research artifact. It beats the retained linear checkpoints on the selected verifier-facing metrics used by the leaderboard. Gemini v5 is negative evidence: it underperformed the gap-weighted linear baseline. The remaining synthetic headroom is small under the current candidate family because v6 already has complete top-5 and top-10 recall, p99 holdout calls of 1, and 9 hard-case top-1 misses across 1,489 winner-bearing hard batches.

Mean winner position falls from 19.995 under exhaustive order to 1.0166 under learned ordering, a reduction of approximately 94.9 percent. Compared with the hand-coded energy baseline, the learned model reduces mean winner position by approximately 25 percent.

Table 2: Retained gap-weighted linear holdout candidate-ordering performance.

Mode	Top-1	Top-5	Top-10	Mean calls	P99	Invalid accepts
Random	0.045	0.245	0.507	10.467	20	0
Hand energy	0.763	0.996	1.000	1.362	4	0
Learned	0.983	1.000	1.000	1.017	2	0
Hybrid	0.983	1.000	1.000	1.017	2	0

Table 3: Top-k sweep on the holdout corpus. Values are checked-stop audit rates.

Mode	$k = 1$	$k = 2$	$k = 3$	$k = 5$	$k = 10$	$k = 25$
Random	0.045	0.091	0.144	0.245	0.507	1.000
Hand energy	0.763	0.919	0.969	0.996	1.000	1.000
Learned	0.983	1.000	1.000	1.000	1.000	1.000
Hybrid	0.983	1.000	1.000	1.000	1.000	1.000

The checked-stop audit is computed offline after suffix labels are known. It measures whether a deterministic suffix certificate would have justified stopping at that k in the same case.

A later suffix-bound benchmark replaces that offline audit with a deterministic early-stop certificate over the unchecked suffix. A checked verifier winner may stop only when it dominates the checked prefix and every unchecked candidate has a deterministic objective upper bound no better than the winner. On the committed bounded synthetic run with seed 20260541, learned and hybrid ordering each achieved mean verifier calls of 1.0084, p99 verifier calls of 1, zero invalid accepts, and zero full fallback cases across 119 evaluated batches. Hand energy averaged 1.4202 verifier calls, and random ordering averaged 13.1849.

The cross-seed suffix-bound stress then repeated the benchmark across seeds 20260541, 20260542, and 20260543 with 20, 32, and 50 candidates per batch.

Table 4: Suffix-bound cross-seed stress.

Mode	Configs	Mean calls	P95 max	P99 max	Fallbacks	Invalid accepts
Exhaustive	9	2.3631	5.0000	7.0000	0	0
Hand energy	9	1.3935	4.0000	6.0000	0	0
Learned	9	1.0132	1.0000	4.0000	0	0
Hybrid	9	1.0132	1.0000	4.0000	0	0
Random	9	17.1010	48.0000	50.0000	16	0

Learned and hybrid ordering kept objective-equivalent acceptance, suffix-stop, and certificate-ok rates at 1.0 across all nine configs. This strengthens the bounded synthetic utility claim while leaving the same candidate-family coverage and real replay obligations.

The adversarial suffix stress then injected high-declared-output invalid candidates into the unchecked suffix after the verifier winner was checked.

The follow-up adversarial family stress generated 944 verifier-invalid suffix cases across 8 invalidity families. Every case was deterministically disqualified, every with-disqualifier suffix certificate passed, and high-declared-output cases still failed under declared-output-only bounds.

This supports a sharper design lesson: suffix certificates need verifier-derived deterministic invalidity signals. Raw declared outputs are too weak against attractive invalid suffixes.

Table 5: Suffix-bound adversarial stress.

Metric	Value
Evaluated batches	119
Adversary invalid count	119
Adversary disqualified count	119
With-disqualifiers certificate ok	119
Without-disqualifiers certificate ok	0
Declared-output-only forced fail	119

Table 6: Suffix-bound adversarial family stress.

Metric	Value
Evaluated batches	118
Adversarial families	8
Total adversarial cases	944
Adversary invalid count	944
Adversary disqualified count	944
With-disqualifiers certificate ok	944
Without-disqualifiers certificate ok	590
Observed disqualifier count	8

9 Langevin Discovery Boundary

The internal Gemini work also introduced Langevin-style candidate refinement:

$$x_{t+1} = x_t - \eta \nabla E(x_t) + \sqrt{2\eta}\epsilon.$$

This is proposal search. It can lower learned energy, but lower learned energy does not imply verifier acceptance. The committed receipt records the critical negative case:

seed_verifier_ok	true
refined_verifier_ok	false
accepted_refinement	false
fallback_to_seed	true
selected_verifier_ok	true.

The implementation selects a refined proposal only when deterministic verification accepts it and the learned energy improves. Otherwise it falls back to the verifier-backed seed when the seed is valid. ZenoGuard remains a soft advisory prior; it is not a validity proof, a settlement authorization rule, or a replacement for the UPBA verifier.

10 AutoTrader Refiner And JEPA Boundary

The AutoTrader refiner applies bounded Langevin-style proposal search to adjustable execution features. The refined feature map is canonicalized and rechecked by deterministic policy labels before selection. The committed baseline receipt records 160 evaluated contexts, 160 accepted refinements, zero selected invalid proposals, mean selected objective gain of 12.0035, and mean selected energy delta of -4.6224. The optimizer replay for the preconditioned refiner records mean selected objective gain of 13.0087, mean selected energy delta of -4.6852, and winning settings `precondition_decay=0.9`, `lr=0.04`. This is hard synthetic evidence. It shows the correct proposal path, while real shadow replay remains required before promotion.

ZenoJEPA and ZenoLogic are also bounded advisory surfaces. The JEPA receipt shows a latent future-tension score ranking a balanced action below a draining action, with tensions 0.3094 and 1.3516 respectively. The same receipt records a ZenoLogic hazard: `EnergyNot` can make a hard-barrier violation lower energy than a valid case. ZenoLogic can compose advisory energies, but it does not create a formal verifier, and `EnergyNot` must not be used over hard safety predicates.

The source-level JEPA/UX pass moves the useful part of future-tension scoring into `src.energy`. The default AutoTrader JEPA model projects candidate features into liquidity, drawdown, execution, and operational fragility latents. The UX helper then turns policy labels, future-tension diagnostics, and suggested controls into compact advisory cards. The committed receipt records 96 generated contexts, future weight 0.1, later-policy-failure AUC 0.8144, slippage, budget, and drawdown stress correlations of 0.6133, 0.5592, and 0.5556, suggested-control best reduction rate of 1.0000, blocked-status match rate of 1.0000, future-warning match rate of 1.0000, zero invalid accepts, balanced future tension of 0.9103, and fragile future tension of 4.7643. The old JEPA-over-hand ordering is recorded as negative evidence: hand+JEPA top-5 recall is 0.8021 on this receipt, while learned+JEPA top-5 recall is 1.0000. Future tension is best used as a risk explanation, warning signal, and counterfactual-control score layered behind learned AutoTraderEnergy. The UX improvement is practical: blocked actions can show the exact deterministic policy guard reason, while policy-valid fragile actions can be shown as risk-review candidates with controls such as refreshing quote receipts, lowering notional, tightening route selection, or waiting for budget recovery. The card is an explanation wrapper around advisory scores and policy labels; it is not an approval.

11 Model Audit

The retained gap-weighted linear checkpoint has no forbidden label-like feature names and no nonzero reserved-feature weights. Its largest positive weights are hard verifier-shaped penalties inherited from hand initialization: negative reserves, CPMM invariant failures, limit-price violations, balance violations, malformed fill vectors, schema/policy mismatches, and output mismatches. Its largest negative weights reward executed volume and surplus:

candidate_normalized_executed_volume	-58.0118
candidate_normalized_surplus	-28.7780
candidate_volume_log1p	-9.7421
candidate_surplus_signed	-7.6317.

This supports the intended interpretation: malformed candidates stay expensive, while higher-objective valid-looking candidates move earlier.

12 Why This Is Not Production Ready

ZenoEnergy v0 is research-useful and operationally small, yet several blockers prevent production readiness.

1. **Synthetic bounded evidence.** The current results come from a synthetic generator. This is appropriate for a controlled benchmark, but it does not show distributional fit to real orderflow.

2. **No privacy-approved replay corpus.** A production-adjacent claim needs non-private or properly redacted replay data with verifier-backed labels.

3. **Candidate-family exactness is still a proof obligation.** The synthetic corpus is an experiment. Production needs exact candidate generation or a certificate that the bounded family under discussion is sound and complete.

4. **UPBA v2 bounded-grid optimality evidence is incomplete.** The scorer helps rank v2 candidates, while production settlement claims still need the v2 bounded-grid optimality verifier or an equivalent certificate.

5. **Live early stopping requires production-grade suffix evidence.** Top-k recall is an empirical accelerator. The current suffix-bound certificate is a deterministic prototype over generated finite lists. The adversarial stress shows that declared-output-only bounds are insufficient and that deterministic disqualifiers are part of the certificate design. The multi-family stress broadens that result across schema/policy, fill coverage, all-zero, limit-price, price-objective, reserve/invariant, and output-mismatch failures. Production early stop still requires candidate-family coverage, real replay, and source-manifested evidence that the bound remains useful on representative traffic.

6. **Scope is narrow.** The experiment covers a single pool, CPMM, exact-in swaps, fixed admitted intents, and partial fills. Multi-pool routing, exact-out, confidential orderflow, latency races, and adversarial live behavior remain outside the measured surface.

7. **Operational controls are missing.** A production deployment needs model artifact signing, version pinning, rollback policy, drift detection, resource bounds, replay conformance, incident response, and governance review.

These are engineering and assurance blockers. They do not undermine the research result. They define the next release gate.

13 Production Gate and Evidence Bundle

ZenoEnergy now has a production-adjacent evidence path:

`tools/build_zenoenergy_production_evidence_bundle.py`.

The bundle composes five deterministic artifacts:

`zenodex/energy/upba_real_replay_report/v1`,
`zenodex/energy/autotrader_real_shadow_report/v1`,
`zenodex/energy/replay_source_manifest_check/v1`,
`zenodex/energy/replay_coverage_profile_check/v1`,
`zenodex/energy/production_promotion_gate/v1`.

The bundle itself uses schema

`zenodex/energy/production_evidence_bundle/v1`.

Operators build source manifests with

`tools/build_zenoenergy_replay_source_manifest.py`.

That command computes canonical source-report hashes, attaches deterministic replay and no-live-secrets attestations, records the secret-scan result, runs the manifest checker, and writes the manifest only when the check passes. Operators can generate the secret-scan report with

`tools/check_zenoenergy_replay_secret_scan.py`.

The scanner writes

`zenodex/energy/replay_secret_scan/v1`,

catches obvious key material and sensitive JSON keys, and can be supplied to the manifest builder with `-secret-scan-report`. Dirty scans and source-count mismatches fail closed. A clean scan is a packaging guardrail; production promotion still requires the production gate and source-manifested real replay reports.

The coverage-profile checker writes

`zenodex/energy/replay_coverage_profile_check/v1`.

It requires UPBA replay breadth across pools, intent-size buckets, candidate families, hard-negative families, and market days. It requires AutoTrader shadow replay breadth across strategy, guard, and decision families. The profile is a breadth guard against aggregate-count-only promotion; it does not prove representativeness of future traffic.

The release predicate is:

$$\begin{aligned}\text{ProductionEvidenceBundle} := {} & \text{UPBARealReplayReport} \\ & \wedge\ \text{AutoTraderRealShadowReport} \\ & \wedge\ \text{ReplaySourceManifestChecks} \\ & \wedge\ \text{ReplayCoverageProfileChecks} \\ & \wedge\ \text{ProductionPromotionGate}.\end{aligned}$$

The production gate requires at least 1,000 real UPBA replay batches, 20,000 real UPBA candidates, 500 real AutoTrader shadow contexts, 5,000 AutoTrader shadow rows, seven market days, top-25 recall at least 0.99, zero invalid accepts, deterministic replay, no live secrets, source manifest checks, and coverage profile checks. It also requires learned mean verifier or guard calls below hand energy.

Malformed evidence fails closed. Well-formed but insufficient evidence produces `decision: blocked`. A passing bundle can only support advisory ranking:

$$\text{LowEnergy}(c) \wedge \text{VerifierRejects}(c) \Rightarrow \text{SettlementRejected}.$$

The current gate remains blocked because the repository contains synthetic and fixture evidence, plus the tooling needed to evaluate real evidence, while the required real replay reports have not yet been supplied.

14 AutoTrader Transfer

The ZenoEnergy pattern applies naturally to AutoTrader if AutoTrader decisions are represented as finite candidate plans. Let $A(s)$ be a candidate action or strategy set for trading state s, let $G_s(a)$ be the deterministic policy, risk, authorization, and budget gate, and let $U_s(a)$ be the deterministic utility or risk-adjusted objective used by the verifier. An advisory scorer can rank candidates:

$$E_\phi(s, a) \to \mathbb{R},$$

while the execution rule remains

$$\text{executable}(a) := G_s(a).$$

This would let AutoTrader use learned energy for ordering candidate strategies, repairing low-quality plans, and reducing evaluation cost. The same safety rules carry over:

- the model ranks strategies but cannot authorize trades;
- deterministic policy gates decide execution;
- model output does not enter balances, positions, state roots, or signed orders;
- full fallback checks a permutation of the original candidate plans;
- early stop requires a deterministic certificate for unexamined candidate plans.

The best first AutoTrader experiment is not live trading. It is a replayed, non-private candidate-plan benchmark with deterministic risk labels. The metric should mirror ZenoEnergy: does learned ordering place the gate-accepted, highest-utility plan in the first few verifier checks without changing any execution predicate?

15 Conclusion

ZenoEnergy v0 provides strong evidence that small energy rankers can reduce candidate-search cost while preserving deterministic settlement verification. The current v6 MLP moves the winner to position 1.0025 on average on the holdout corpus and keeps top-5 and top-10 recall complete across the measured lanes. The result is useful because the model is still small, CPU-friendly, and isolated from consensus-critical paths.

The UPBA ranker is near a synthetic-distribution plateau for the current candidate family. Further UPBA work should target the remaining valid-vs-valid misses, production-shadow replay, and suffix-certificate utility rather than raw parameter scaling. The release path stays the same: keep the scorer advisory, add real replay evidence, finish exact candidate-generation and v2 optimality certificates, require a deterministic suffix certificate for live early stop, and pass the production evidence bundle gate. AutoTraderEnergy now deserves the larger share of modeling work because its synthetic gains are large, its checked refiner has the right policy-gated shape, and its real shadow evidence is still missing. The next UX milestone is real shadow replay that verifies cards surface the correct guard reason, risk level, future-tension warning, and operator control without hiding the deterministic authority boundary.

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